

WM571-B/C
UV254 COD Sensor

Operation Manual

Table of Contents

Introduction	3
1. Product Description	4
Test Principle	4
Sensor Features	4
1.1 Product Introduction	5
1.2 Cable Definition	6
1.3 Technical Specifications	7
2. Installation	8
2.1 Configuration	8
2.2 Installation Precautions	8
2.3 Installation Steps	9
3. Calibration	11
3.1 Brief Description	11
3.2 Standard Solution Preparation	11
4. Maintenance Schedule and Method	13
4.1 Maintenance Schedule	13
4.2 Maintenance Method	13
4.3 Cleaning Wiper Replacement	15
5. Frequently Asked Questions	16
6. Quality Assurance Explanation	17

Introduction

Dear customers,

Thank you very much for purchasing our COD sensor. Before you use it, please read this manual carefully, it will be of great help to the use and maintenance of this instrument, and can avoid unnecessary troubles due to improper operation and maintenance.

Please follow the operating procedures and precautions of this manual.

In order to ensure that the after-sales protection provided by this instrument is effective, please do not use and maintain this instrument by methods other than those specified in this manual.

Any failures and losses caused by non-compliance with the precautions specified in this manual are not covered by the manufacturer's warranty, and the manufacturer does not assume any related responsibilities. Please keep all documents in a safe place. If you have any questions, please contact our after-sales service department.

When you receive the instrument, please carefully open the package and check whether the instrument and accessories are damaged due to transportation. If any damage is found, please contact our company's after-sales service department, and save the packaging so that it can be sent back for processing.

When the instrument fails, please do not repair it yourself, please contact our after-sales service department.

1. Product Description

This UV254 COD sensor is based on the principle of ultraviolet absorption method, combined with a lot of application experience, has carried out a number of upgrades and innovations on the basis of the previous generation, not only smaller in size, but also integrated with the original split cleaning wiper, so that it is more convenient to install and operate, and has higher reliability.

This UV254 COD sensor does not require reagents, is pollution-free, and is more economical and environmentally friendly. It can monitor water quality online for a long time. A turbidity detection light source is integrated inside the sensor, which can automatically compensate for the turbidity interference of the water to be measured. The sensor has an automatic cleaning wiper device, which still has excellent stability even for long-term monitoring.

Test Principle

Many organic substances dissolved in water absorb ultraviolet light. Therefore, the total amount of organic pollutants in water can be measured by measuring the degree of absorption of these organic substances to 254nm wavelength ultraviolet light. The WM571-B/C sensor uses two light sources, one for 254nm ultraviolet light and one for turbidity compensation, which can automatically eliminate water turbidity interference, thereby achieving more stable and reliable measurement.

Sensor Features

- Digital sensor, digital RS-485 output, Modbus protocol;
- No reagents required, no pollution, more economical and environmentally friendly;
- Can measure parameters such as COD, TOC, turbidity and temperature;

- Automatically compensates for turbidity disturbances for excellent test performance;
- With self-cleaning wiper to prevent biofouling and longer maintenance intervals.

1.1 Product Introduction

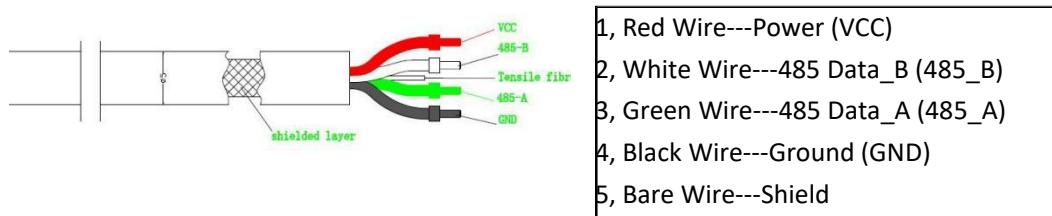


▲WM571-B Self-cleaning COD Sensor Appearance and Dimensions (Standard range)



1.2 Cable Definition

4-wire AWG-24 or AWG-26 shielded wire. OD=5~5.5mm



1.3 Technical Specifications

Name	UV254 COD Sensor	
Model	WM571-B	WM571-C
COD Range	0-500 mg/L equiv.KHP	0-1500 mg/L equiv.KHP
COD Accuracy	$\pm 5\%$	
COD Resolution	0.01 mg/L	
TOC Range	0-180 mg/L	0-600 mg/L
TOC Resolution	0.01mg/L	
Turbidity Range	0-300 NTU	0-500 NTU
Turbidity Accuracy	$\pm 5\%$ or 0.5NTU	$\pm 5\%$ F.S.
Turbidity Resolution	0.01NTU	
Temperature Range	0~50°C (no freezing)	
Housing IP Rating	IP68	
Maximum Pressure	3 bar	
User Calibration	1 point or 2 points	
Power Requirements	DC 12V -24V, current<50mA (no cleaning wiper); current < 200mA (when the cleaning wiper is activated)	
Sensor Interface	RS485, Modbus protocol	
Sensor Dimension	$\Phi 50\text{mm} \times 179\text{mm}$	
Cable Length	10 meters (default), customizable	
Shell Material	316L (titanium alloy can be customized)	

Note: The above technical parameters are the data in the laboratory standard solution environment; Sensor life and maintenance calibration frequency are related to actual field conditions.

2. Installation

Note

Spread out the cables before proceeding.

2.1 Configuration

Standard Configuration	Quantity	Remark
COD Sensor	1	Including hoisting sheet metal parts, hoop
Metal Protective Cover	1	
Cable	1	
Wiper Component	1	

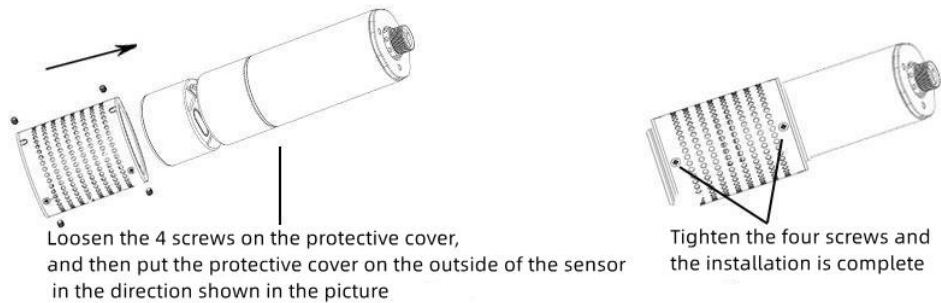
2.2 Installation Precautions

- ① When hanging the sensor, avoid the sensor hitting the wall or other water conservancy facilities caused by the water flow. If the water flow is strong, fix the sensor.
- ② Consider water level fluctuations, submerge the sensor below the lowest possible water level of 30CM.
- ③ The sensor is installed in a position where there are no air bubbles in the water, as far away as possible from the aeration port.
- ④ Considering the specific working conditions such as the deposition of impurities in the water body, it is recommended to install the sensor horizontally and keep the measurement window vertically downward.

2.3 Installation Steps

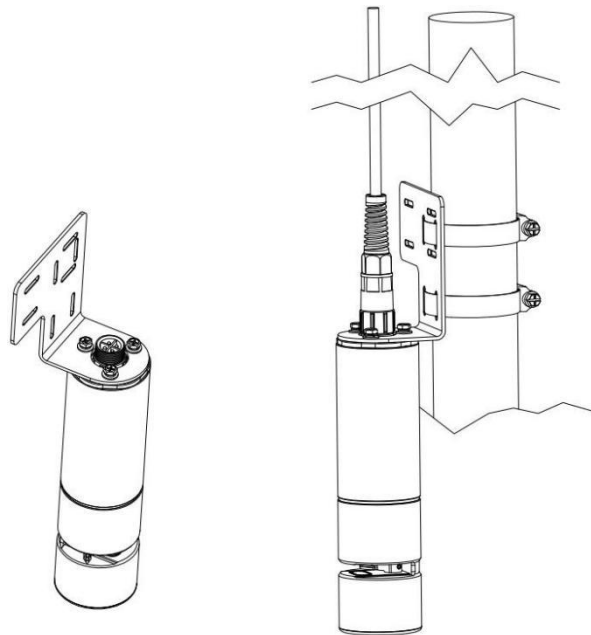
(1) Protective cover installation

After taking out the sensor, you need to install a protective cover on the sensor, as shown in the picture below (the 4 screws on the cover need to be tightened, pay attention to the 4 screws on the top, bottom and diagonal, and the other 4 screws do not need to be tightened).

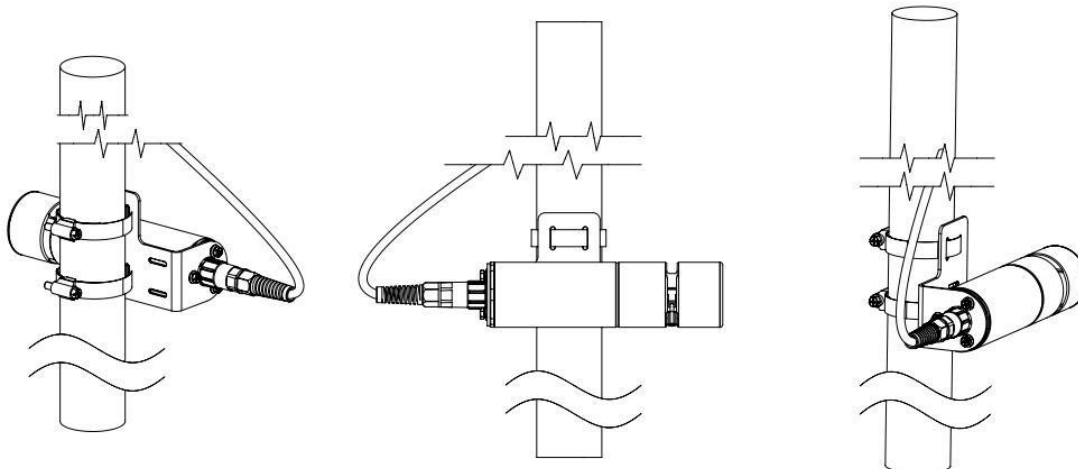


(2) Field fixed installation

It is recommended to carry out fixed installation according to the following two ways (1) and (2) in the schematic diagram.



Schematic diagram (1) The hoisting method is vertical hoop type, suitable for environments with no rapids and few water debris



Schematic diagram (2) The hoisting method is horizontal hoop type, suitable for water with many debris and turbidity



Warning

- (1) Please install the protective cover correctly.
- (2) Please do not use the sensor cable to lift the sensor.
- (3) Do not cover the measuring surface with the lifting accessories.

3. Calibration

3.1 Brief Description

The COD sensor supports zero-point self-calibration, one-point and two-point calibration, customers can also develop their own calibration functions according to the Modbus communication protocol.

3.2 Standard Solution Preparation

(1) Required equipment and raw materials

① Equipment

Analytical balance, spatula, 50mL beaker, 100mL graduated cylinder, glass rod, pipette, 1L volumetric flask.

② Raw materials

KHP (Potassium hydrogen phthalate, $C_8H_5KO_4$), CAS# 877-24-7, as a commonly used stain in environmental research, it can be used for COD calibration, deionized water.

(2) Preparation method

① Weigh 1.2754g KHP and dissolve in deionized water, constant volume to 1L, get 1500mg/L COD mother liquor;

② According to the required concentration, dilute the solution in step ①. For example: measure 100mL of the ① standard solution and dissolve it in deionized water, and dilute to 1L to obtain a 150mg/LCOD standard solution.

③ Other concentration standard solutions can be calculated and prepared according to the dilution method.



Notice

The standard solution is best prepared and used immediately to ensure accuracy;

High-concentration mother liquor (such as 1500mg/L) can be sealed and stored in the refrigerator for subsequent dilution; the storage period is recommended not to exceed one month.

4. Maintenance Schedule and Method

4.1 Maintenance Schedule

Although the WM571-B/C is equipped with a self-cleaning wiper as standard, but the harsh working conditions will still cause the sensor to be stained. In order to ensure accurate measurement, cleaning is very important. Regular cleaning of the sensor will help the stability of the data.

Maintenance Tasks	Recommended Maintenance Frequency	
Sensor cleaning	It is recommended to clean once every 3~4 weeks	It is recommended to clean once every 2~3 weeks
Sensor calibration	According to the working conditions and user needs, but no later than once every 3 months	
Maintain and inspect self-cleaning wiper	Replace the wiper with a new one every 3~6 months (depending on the specific working conditions). Return to factory for replacement dynamic seals every 18 months.	Replace the wiper with a new one every 2 months (depending on the specific working conditions). Return to factory for replacement dynamic seals every 18 months.

4.2 Maintenance Method

(1) Sensor outer surface cleaning: Wash the outer surface of the sensor with tap water, and if debris remains, wipe it with a damp soft cloth. For some stubborn dirt, you can add some household detergent to tap water to clean.

(2) Sensor measuring window cleaning: Light window scrubbing with a fine-bristle brush or cotton swab, avoid scratching with sharp objects to avoid scratching the light window.

(3) Check sensor cable: The cable should not be tight during normal work, otherwise the wire inside the cable may be broken for a long time, causing the sensor to work

abnormally.

(4) Check the cleaning wiper: Check whether the wiper is in effective contact with the light window, whether it rotates normally, and whether it is loose; if the wiper is severely worn and cannot scrape the window surface, it needs to be replaced, if the wiper is loose, it needs to be retightened.

(5) After 18 months of continuous use, it needs to be returned to the factory to replace the dynamic sealing device.



Precautions

(1) Sensor contains sensitive optics and electronics. Make sure the sensor is not subject to mechanical impact. No user-maintainable parts inside the sensor.

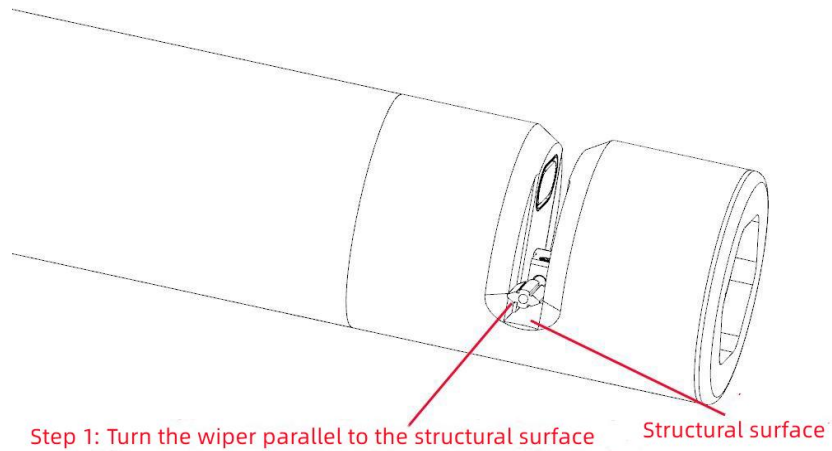
(2) The sensor self-cleaning wiper has a geared motor inside, do not use external force to rotate the cleaning wiper under any circumstances (except for changing the cleaning wiper against the wall), or hinder the rotation of the cleaning wiper. Larger external force factors may cause damage to the geared motor.

(3) If there are many sundries in the water body at the installation point, it is recommended to install a protective net or a protective sleeve around the sensor to prevent the sundries in the water from jamming the cleaning wiper.

(4) The sensor installation should avoid facing the water flow and the location with more air bubbles.

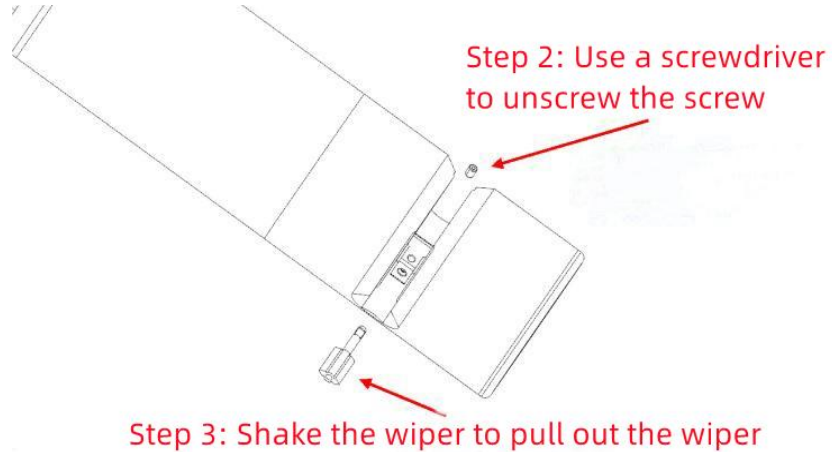
4.3 Cleaning Wiper Replacement

(1) The sensor is placed horizontally, and the wiper is turned to be parallel to the structural surface;

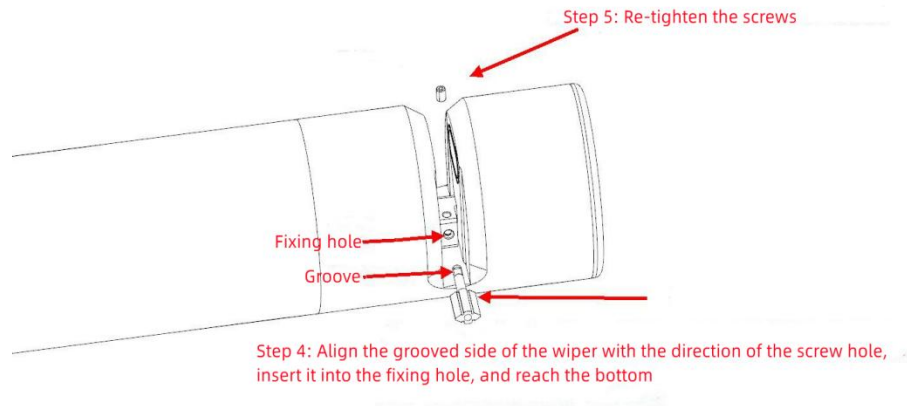


(2) Using a screwdriver, unscrew the screw;

(3) Shake the wiper to pull out the wiper;



(4) Align the grooved side of the wiper with the direction of the screw hole, and insert it into the fixing hole to the bottom;



(5) Re-tighten the screws.

5. Frequently Asked Questions

Table 5-1 lists possible COD sensor problems and their solutions, if your problem is not listed or the solution does not solve your problem, please contact us.

ERROR	POSSIBLE CAUSE	SOLUTION
Communication failure	Power supply or wiring error	Check whether the power supply and wiring are correct according to the instruction
	Interface or protocol issues	1、 Use our upper computer software to check whether the communication is normal. 2、 Check according to the supporting communication protocol of the product.
No change in reading	Cleaning wiper failure	Check whether the wiper is entangled by debris, if so, please remove them gently. Turn on the power again and observe whether the wiper rotates. If it cannot rotate or rotates abnormally, please contact customer service.
		Check whether the power supply meets the requirements, avoid the wiper not working properly due to low power supply.
	Hardware or software Failure	Contact customer service.

Reading is too high, too low or unstable	The sensor is heavily soiled	Clean the sensor body, especially the light window surface
	Sensor cleaning wiper are highly worn	Replace the cleaning wiper
	Calibration required	Perform user calibration
Other errors	Please contact our after-sales department	

Table 5-1 Frequently Asked Questions List

6. Quality Assurance Explanation

(1) The warranty period of the whole sensor is 1 year (consumables such as wiper and adapter are not included).

(2) This warranty does not cover the following situations:

- ① Damage caused due to force majeure, natural disasters, social unrest, war (published or unpublished), terrorism, civil war or any government forced;
- ② Damage caused due to improper use, negligence, accident, or caused by the improper application and installation;
- ③ Freight for the product shipped back to our company;
- ④ Freight for parts or products express or delivery within the warranty;
- ⑤ Travel expense for repair in local in warranty.

(3) The quality assurance includes all content of products provided by us.

① This warranty constitutes a final, complete and exclusive statement about the terms of the warranty. No person or agent is authorized to formulate other warranties in the name of our company.

② The above-mentioned remedial measures such as repair, replacement or refund of payment are special cases that do not violate this warranty, and remedial measures such as replacement or refund of payment are all aimed at the company's product itself. Based

on strict liability obligations or other legal theories, our company is not liable for any other damage caused by product defects or due to negligence in operation, including the case of subsequent damages that are causally related to these situations.